

Saiakhil Chilaka

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EDUCATION

Cornell University, College of Arts & Sciences

Ithaca, NY

B.A. in Computer Science

Expected May 2028

- **Relevant Coursework:** Distributed Systems for Machine Learning, Intro Machine Learning, Systems, Data Structures

EXPERIENCE

Cornell Generative AI Lab

August 2026 – Present

Applied AI Researcher

Ithaca, NY

- Researching context management systems for long horizon agents to mitigate entity binding failures & governance decay

MotherDuck

May 2026 – August 2026

Software Engineering Intern

New York, NY

- Built the internal evaluation harness to benchmark AI agents across 40 MotherDuck tasks over the CLI & MCP
- Examined failed agent trajectories to write CLI guides for 3 features, taking task category accuracies from ~16% to 100%
- Reduced agent token cost on the MotherDuck CLI up to 57% through A/B testing 5 output design principles

Vanasse Hangen Brustlin, Inc.

January 2026 – May 2026

Software Engineering Intern

Remote/Raleigh, NC

- Designed an agentic PDF-to-form pipeline for car crash reports, cutting analyst time by 66% over 10k+ crashes annually
- Engineered an automated geocoder & RAG classification layer w/ project docs as context, achieving ~90% accuracy

Duke University

August 2024 – December 2024

Research Intern

Durham, NC

- Published first-author IEEE paper on CNN ensemble for Multiple Sclerosis diagnosis; 97.5% accuracy & 97.8% AUC-ROC
- Fine-tuned VGG19/ResNet50 layers via depth-wise separable convolutions, reducing model size 14x for edge deployment
- Injected Gaussian white noise alongside rotation, shear, and zoom augmentation to improve generalization by up to 7%

North Carolina State University

February 2024 – September 2024

Research Intern

Raleigh, NC

- Implemented knowledge distillation on a skin cancer classifier via temperature-scaled KL divergence + cross-entropy loss, lifting accuracy 19% & bias 37% while compressing model size 18x & maintaining sub-10s inference on Raspberry Pi 4
- Trained a generative adversarial network to synthesize training data, improving overall performance score by 13%

PROJECTS

Pass@k is Not a Property of a Model | *Python, PyTorch, verl, AWS, SkyPilot*

- Showed central results of *Limit of RLVR* (Yue et al.) fall 3x (36 to 12 pts) by altering the wording of the test questions
- Evaluated that an ensemble of 3 independently trained Qwen2.5 runs beats one model by 6% at matched sample budget
- Showed the main reasoning benchmark (MATH-500) is 98.8% solved by base models it tests, leaving 6/500 trials usable

Agentic Polymarket Predictor | *Python, Neo4j, Qdrant, Claude API, FastAPI, Modal*

- Built a 5-stage agentic predictor for Polymarket, achieving 30.5% out-of-sample ROI on 950+ markets and 140+ trades
- Architected a hybrid RAG pipeline over Neo4j & Qdrant via Reciprocal Rank Fusion with a BGE reranker
- Beat or matched the market on Brier scores on the 53% of markets the model chose, validated across multiple evaluations

TECHNICAL SKILLS

Languages: Python, TypeScript, R, SQL, Java

Frameworks & Libraries: PyTorch, TensorFlow, scikit-learn, LightGBM, verl, FastAPI, HuggingFace Transformers

Databases & Infrastructure: PostgreSQL, Redis, Supabase, Neo4j, Qdrant, Celery, Modal, AWS ECS, Docker, Git

ML/AI: MCP, RAG, Agent Evaluation, LoRA, Knowledge Distillation, Reinforcement Learning, Reranking, Vector Search

PUBLICATIONS

Chilaka et al. "Pass@k Is Not a Property of a Model," Preprints 2026, 2026081495, doi: 10.20944/preprints202608.1495.v1

Chilaka et al. "CNN Ensemble MS Lesions Diagnosis for Embedded Systems," *IEEE*, doi: 10.1109/MCSoc64144.2024.00079